Safety Zone Detection — Implementation Plan

Overview

Add real-time **"Safe / Caution / Unsafe"** location safety assessment for every tracked traveler. The assessment is computed server-side, stored in Supabase, and displayed in:

1. **Mobile app** (the traveler themselves) — a live safety status banner on the Active Tracking screen
2. **Web dashboard** (trackers / admin) — safety badge on device cards, the bottom panel, map markers, and a new danger-zone heatmap overlay

Architecture

What gets built

1. Supabase Migration — `safety_assessments` table

New table storing the latest safety assessment per user:

- `id`, `user_id`, `travel_card_id`, `lat`, `lon`
- `safety_status` enum: `safe`, `caution`, `unsafe`
- `risk_score` (0.0 = perfectly safe → 1.0 = extremely dangerous)
- `risk_factors` JSONB array (each factor with `type`, `label`, `severity`, `distance_km`)
- `nearby_hazards` JSONB (crime reports, isolation, hazard zones, emergency distance)
- `assessed_at` timestamp
- RLS: users can read their own; admins can read all

2. Server-side Safety Assessment Endpoint (`/api/assess-safety` on server.py)

The Python server gets a new endpoint that:

1. Queries Overpass API for the area around the user's coordinates
2. Checks for **police stations / hospitals / fire stations** — distance = emergency access factor

3. Checks for **isolation** — absence of roads, buildings, populated areas in radius
4. Checks for **official hazards** — flood zones, restricted areas, landslide-prone areas via natural=* tags
5. Computes a composite `risk_score` and `safety_status`
6. Writes results to `safety_assessments` table via Supabase

3. Mobile App — Safety Status Banner

New `SafetyBanner` component on the Active Tracking screen showing:

- Color-coded status: 🟢 Safe / 🟡 Caution / 🔴 Unsafe
- List of active risk factors with icons
- "Heading towards unsafe area" predictive warning using movement vector

4. Web Dashboard — Safety Integration

- **DeviceCard**: Safety status dot next to freshness badge
- **DevicePanel**: Full safety assessment with risk factors
- **MapView**: Danger zone circles on the map around detected unsafe areas
- **New `useSafetyAssessments` hook**: Real-time subscription to `safety_assessments`

Files to create/modify

File	Action	Description
`supabase/migrations/add_safety_assessments.sql`	**Create**	New table + RLS
`server.py`	**Modify**	Add `/api/assess-safety` endpoint
`LocationShare/lib/safetyAssessment.js`	**Create**	Safety assessment fetcher for mobile
`LocationShare/components/SafetyBanner.js`	**Create**	Safety status UI component
`LocationShare/screens/ActiveTracking.js`	**Modify**	Integrate SafetyBanner
`WebDashboard/.../hooks/useSafetyAssessments.js`	**Create**	Realtime hook for safety data
`WebDashboard/.../utils/safetyHelpers.js`	**Create**	Safety status helpers & colors
`WebDashboard/.../components/SafetyBadge.jsx`	**Create**	Reusable safety status badge
`WebDashboard/.../components/SafetyPanel.jsx`	**Create**	Detailed safety breakdown panel
`WebDashboard/.../components/DeviceCard.jsx`	**Modify**	Add safety badge
`WebDashboard/.../components/DevicePanel.jsx`	**Modify**	Add safety info section
`WebDashboard/.../components/MapView.jsx`	**Modify**	Add danger zone circles
`WebDashboard/.../App.jsx`	**Modify**	Wire up safety hook + panel
`WebDashboard/.../index.css`	**Modify**	Safety-related styles